QUISQUISLINGO - ANDROID GUIDE

ABOUT QUISQUISLINGO

QuisquisLingo is an offline-first language-learning application for learners and course creators.

It provides structured language courses made of Lessons, GuideBooks, Rounds, interactive exercises, review activities and Duels. Depending on the course, exercises may use text, images, recorded audio and text-to-speech.

QuisquisLingo also includes authoring tools for creating, editing, inspecting, importing and exporting language courses. It is therefore intended both for people who want to study a language and for authors, teachers and other users who want to create their own learning material.

Learner progress, profiles, settings and locally created content are stored locally. QuisquisLingo does not require a user account or a permanent server connection for its normal learning functions.

QuisquisLingo is open-source software. The software source is licensed under the Mozilla Public License 2.0. Courses, images, audio and other content may have their own separate licences or rights.

Official source repository:

```
https://github.com/Quisquisnaut/QuisquisLingo
```

ABOUT THE INITIAL ANDROID VERSION

The first publicly distributed Android package of QuisquisLingo is intended primarily for testing and evaluation.

It is initially distributed as a DEBUG APK rather than as a production release build.

A debug APK is a complete installable Android application, but it differs from a final production package in several important ways.

A debug build:

- is intended for development and testing;
- is signed with development/debug signing credentials rather than a final production signing key;
- contains debugging support;
- may be larger than a production build;

- may use more resources or perform less efficiently than an optimized release build;

- is not intended for publication through Google Play as the final production application;

- may produce additional diagnostic information useful during testing.

The debug status does NOT mean that the application is incomplete or that it must be run from Android Studio.

The supplied APK can be copied directly to a compatible Android phone or tablet and installed as a normal application, subject to the security policy of that device.

A later QuisquisLingo Android distribution may use a separately signed release APK or Android App Bundle.

THE APK FILE

The initial test package may be supplied as:

```
app-debug.apk
```

or under a QuisquisLingo-specific distribution filename.

If you build the debug APK yourself with Flutter, the default output is normally:

```
build/app/outputs/flutter-apk/app-debug.apk
```

The APK is the application installation package.

Unlike the Windows or Linux packages, the APK does not need to be extracted. Do not unzip it before installation.

INSTALLING THE APK ON AN ANDROID DEVICE

The APK can be transferred to the Android device using, for example:

- USB;
- a cloud-storage service;
- email or messaging;
- a web download;
- another file-transfer method.

Once the APK is on the device:

1. Open the APK using the Files application, browser, cloud application or other application from which it was received.

2. Android may ask for permission to install applications from that particular source.

3. If requested, allow that source to install this APK.

4. Return to the APK and confirm installation.

5. When installation is complete, open QuisquisLingo from the Android application launcher.

Android calls this type of installation sideloading because the application is being installed directly rather than through Google Play.

The exact wording and location of the installation permission varies between Android versions and device manufacturers.

Only enable installation permission for a source that you trust.

ANDROID SECURITY WARNINGS

Because the initial QuisquisLingo Android APK is a directly distributed debug build and is not installed through Google Play, Android or Google Play Protect may display additional security or verification messages.

These messages do not by themselves indicate that QuisquisLingo has malfunctioned.

Read every Android security message before proceeding and install the APK only if you obtained it from the intended QuisquisLingo distribution source.

Some devices may prevent installation entirely because of:

- corporate or school management policies;
- parental or administrative restrictions;
- device security configuration;
- restrictions on installation from external sources;
- other manufacturer or Android security policies.

QuisquisLingo cannot override these device policies.

MANAGED AND WORK DEVICES

An Android device managed by an employer, school or other administrator may restrict or completely disable sideloading.

The APK may therefore work on an ordinary Android device but be refused on a managed device.

If installation is blocked by device-management policy, the restriction must be changed by the device administrator. QuisquisLingo does not attempt to circumvent Android management or security controls.

UPDATING A DEBUG INSTALLATION

Android requires application updates to be signed compatibly with the application already installed.

During the initial debug-testing period, this has an important consequence.

If two QuisquisLingo debug APKs are produced using different debug signing keys, Android may refuse to install the new APK over the existing one.

In that case it may be necessary to uninstall the existing QuisquisLingo installation before installing the new APK.

WARNING:

Uninstalling an Android application normally removes that application's local private data.

Before uninstalling QuisquisLingo, export or back up any learner or course data that you need to preserve using the application's available backup and export functions.

Once QuisquisLingo moves to a stable production signing process, maintaining the same signing identity between releases will be important for normal in-place updates.

SYSTEM REQUIREMENTS

The Android package requires:

- an Android device compatible with the Android version requirements encoded in the supplied APK;
- sufficient free storage for the application and its local data;
- sufficient memory to run a modern Flutter application;
- working audio output if audio exercises or text-to-speech are used.

Android itself checks application compatibility during installation. If the device's Android version is below the minimum supported by the APK, installation will be refused.

An Internet connection is not required for normal offline learning.

Features that deliberately access external web resources naturally require an Internet connection.

Android Studio, Flutter, Dart and development tools are NOT required to install or use a supplied APK.

TEXT-TO-SPEECH ON ANDROID

On Android, QuisquisLingo uses the device's Android text-to-speech facilities.

Available languages and voices therefore depend on:

- the Android version;
- the text-to-speech engine installed on the device;
- the speech languages and voices installed for that engine;
- device-manufacturer configuration.

QuisquisLingo provides learner audio controls under:

```
Settings > Audio Settings
```

These include audio-exercise settings, text-to-speech and voice preference.

Test Voice speaks only the text entered by the user and uses the speech language configured by the selected course.

It does not translate the entered text or infer a language from the text itself.

If a required voice is unavailable, check the Android text-to-speech settings and install an appropriate language or voice if the device supports it.

Text-to-speech is optional. If TTS is unavailable or disabled, QuisquisLingo can still be used, although exercises that require unavailable audio may be excluded according to the current audio settings and course configuration.

FILES AND APPLICATION DATA

QuisquisLingo stores learner data, preferences and other application state locally on the Android device.

Android isolates application-private files from ordinary user-accessible storage.

Some QuisquisLingo operations deliberately allow the user to import or export files through Android's storage and file-selection interfaces.

Depending on the feature being used, QuisquisLingo may create or access:

- learner backup files;
- course export files;
- course import files;

- diagnostic exports;
- images and other course resources selected by the user.

The exact physical Android storage path can vary according to Android version, device manufacturer and the storage API used.

Users should normally use QuisquisLingo's own Import, Export and backup functions rather than attempting to edit internal application directories manually.

LEARNER DATA AND PRIVACY

QuisquisLingo is designed as an offline-first application.

Learner profiles, progress, course state and settings remain on the device during normal use.

Normal learning does not require:

- a QuisquisLingo account;
- cloud synchronization;
- a QuisquisLingo server connection.

QuisquisLingo does not automatically upload its crash or diagnostic logs.

Files leave the device only when a user deliberately uses an applicable sharing, export or external-resource function.

CRASH AND DIAGNOSTIC INFORMATION

QuisquisLingo includes application-level crash and diagnostic logging.

The crash logger can record technical information such as:

- date and time;
- QuisquisLingo version;
- operating-system information;
- device/runtime architecture information;
- processor information;
- locale;
- runtime information;
- error information;
- stack traces when available.

The QuisquisLingo crash logger is designed not to record ordinary learner content such as course answers or profile names.

QuisquisLingo also maintains diagnostic information for selected application errors and technical events.

Settings > Debug provides diagnostic functions where available.

On Android, application-private logs may not be directly visible through an ordinary file manager. Use QuisquisLingo's available diagnostic/export functions when possible.

ANDROID SYSTEM LOGS AND ADB

During testing, developers can obtain additional Android runtime information using Android Debug Bridge, normally called ADB.

ADB is part of the Android SDK Platform Tools.

With a device connected and USB debugging enabled, check the connection with:

```
adb devices
```

Android system and application messages can be viewed with:

```
adb logcat
```

To save a diagnostic log to a file on the development computer:

```
adb logcat > android_logcat.txt
```

Press Ctrl+C when enough information has been collected.

For a more focused debugging session, clear the previous Logcat buffer before reproducing a problem:

```
adb logcat -c
adb logcat
```

Then reproduce the problem and save the relevant output.

Android Logcat is an operating-system diagnostic facility. It can contain messages produced by Android, device services and other software, not only QuisquisLingo.

Review a Logcat file before sharing it.

INSTALLING WITH ADB

Developers and testers can also install the APK directly from a development computer.

First verify that the device is visible:

```
adb devices
```

Then install the APK:

```
adb install path_to_apk
```

To replace an already installed compatible build while preserving its application data when Android permits it:

```
adb install -r path_to_apk
```

An update can still be rejected if the existing and new APKs have incompatible signatures.

USB debugging must be enabled on the Android device before ADB can communicate with it.

TESTING WITH AN ANDROID EMULATOR

QuisquisLingo can also be tested using an Android Emulator.

With a running emulator detected by Flutter:

```
flutter devices
```

A development build can be compiled, installed and started with:

```
flutter run -d emulator-5554
```

Replace `emulator-5554` with the device identifier actually shown by `flutter devices`.

An emulator is useful for development and interface testing, but it does not perfectly reproduce every physical Android device.

QuisquisLingo Android testing should therefore include at least one real Android phone or tablet in addition to emulator testing, especially for:

- text-to-speech;
- recorded audio;
- file import and export;

• device storage;
• small-screen layouts;
• Android lifecycle behaviour.

DEBUG BUILD PERFORMANCE

The initial distributed Android APK is a debug build.

A debug build is not optimized in the same way as a production release build.

As a result:

• startup may be slower;
• some operations may be slower;
• application size may be larger;
• memory and CPU use may differ from a release build;
• development assertions and debugging facilities may be present.

Performance observations made with the initial debug APK should therefore not automatically be treated as representative of the future production release.

Functional problems should still be reported.

REPORTING A PROBLEM

When reporting an Android problem, useful information includes:

• phone or tablet manufacturer and model;
• Android version;
• QuisquisLingo version;
• whether the APK is a debug or release build;
• whether the device is personally managed or organization-managed;
• what you were doing immediately before the problem;
• whether the problem can be reproduced;
• the complete Android error message, if one appeared;
• screenshots when useful;
• QuisquisLingo diagnostic information;
• relevant ADB Logcat output when available.

For audio or TTS problems, also report:

• whether ordinary device audio works;
• which Android TTS engine is selected;
• whether the required speech language is installed;
• whether QuisquisLingo's Test Voice works.

TROUBLESHOOTING

1. Confirm that the APK was downloaded or copied completely.

2. Open the APK and allow installation from that source if Android requests permission and you trust the source.

3. If Android reports that the application cannot be installed, check whether an older QuisquisLingo build is already installed.

4. If an older build exists and the new APK has a different debug signature, back up important QuisquisLingo data before uninstalling the previous build.

5. If the device is managed by an employer, school or administrator, check whether sideloading is prohibited by policy.

6. If QuisquisLingo installs but does not start correctly, restart the application and, if possible, collect ADB Logcat output while reproducing the problem.

7. If text-to-speech does not work, verify the Android TTS engine and installed speech languages.

8. If recorded audio does not work, confirm that normal media playback works on the device and that the device volume is not muted.

9. If import or export fails, verify that the required file or storage permission has been granted when Android requests it.

10. When testing a debug build, distinguish functional failures from general performance differences associated with debug mode.

BUILDING QUISQUISLINGO FROM SOURCE

The complete QuisquisLingo source code is publicly available on GitHub:

```
https://github.com/Quisquisnaut/QuisquisLingo
```

Developers can build the Android application directly from source.

Typical requirements include:

- Git;
- a compatible current Flutter SDK;
- Java as required by the Flutter/Gradle toolchain;
- the Android SDK;

- Android SDK Platform Tools;
- the Android SDK components required by the current project.

Android Studio is a convenient way to install and manage the Android SDK and emulators, but the actual Flutter build can be executed from a terminal once the required toolchain is configured.

Check the development environment with:

```
flutter doctor
```

A typical source preparation sequence is:

```
git clone https://github.com/Quisquisnaut/QuisquisLingo.git
cd QuisquisLingo
flutter doctor
flutter pub get
flutter analyze
flutter test
```

BUILDING A DEBUG APK

To create the same general type of APK used during the initial Android testing period:

```
flutter build apk --debug
```

The resulting file is normally:

```
build/app/outputs/flutter-apk/app-debug.apk
```

This APK can be copied to a compatible Android device and installed for testing.

RUNNING DIRECTLY ON A DEVICE

With a compatible Android device or emulator connected:

```
flutter devices
```

Then:

```
flutter run -d DEVICE_ID
```

For example:

```
flutter run -d emulator-5554
```

Flutter builds the development application, installs it on the selected device and starts it.

FUTURE RELEASE BUILDS

A production-oriented Android APK can be built with:

```
flutter build apk --release
```

The default output is normally:

```
build/app/outputs/flutter-apk/app-release.apk
```

For Google Play distribution, Android applications are commonly distributed using an Android App Bundle:

```
flutter build appbundle --release
```

which normally produces:

```
build/app/outputs/bundle/release/app-release.aab
```

A true public production release also requires an intentional, persistent release-signing configuration.

A debug signing key should not be treated as the permanent identity of a production Android application.

Future QuisquisLingo Android release procedures should preserve the production signing identity so that users can update the application without uninstalling it.

MISSING PLATFORM RUNNER FILES

If a particular source archive does not contain all standard Flutter platform runner files, the QuisquisLingo repository includes platform-preparation scripts.

On Windows:

```
tools\\prepare_flutter_platforms.ps1
```

On Linux or macOS:

```
chmod +x tools/prepare_flutter_platforms.sh
./tools/prepare_flutter_platforms.sh
```

These scripts prepare standard Flutter platform host files without replacing QuisquisLingo's Dart source, courses or application assets.

SOURCE CODE AND LICENCE

Official repository:

```
https://github.com/Quisquisnaut/QuisquisLingo
```

The QuisquisLingo software source is licensed under the Mozilla Public License 2.0.

Courses, images, audio material, Image Bank content and other resources may have separately stated licences, attribution requirements or rights.

Refer to the licence and attribution information supplied with QuisquisLingo and with individual content packages.

SECURITY NOTES

Install QuisquisLingo APK files only from a source you trust.

A directly distributed APK bypasses the normal Google Play installation channel, so the user is responsible for deciding whether to trust the package.

Do not disable Android security features globally simply to install QuisquisLingo.

If Android requires permission to install from an external source, grant it only to the specific application being used to open the trusted APK, and revoke it later if desired.

Do not install modified QuisquisLingo APKs from unknown third-party sources.

IMPORTANT

The initial Android package is a TESTING DEBUG BUILD.

It should be evaluated primarily for functionality, compatibility and bug discovery rather than final production performance.

QuisquisLingo Android behaviour can vary according to Android version, device manufacturer, installed text-to-speech engine, storage implementation, device management policy and hardware configuration.

Testing on an emulator is useful, but testing on at least one physical Android device is strongly recommended before the Android build is considered ready for normal release distribution.